

Introduction to Abstract Algebra II: Homework 4

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Exercise 28.8. Find all the generators of the cyclic multiplicative group of units $\mathbb{Z}_{23}^\times$.

Solution. The group $\mathbb{Z}_{23}^\times$ consists of elements such that $1 \leq a \leq 23$ and $\gcd(a, 23) = 1$. Since 23 is a prime, $\mathbb{Z}_{23}^\times$ consists of all the elements from 1 to 22. The order of $\mathbb{Z}_{23}^\times$ is $\varphi(23) = 22$. The generators of a cyclic group of order n are the elements of order n . The order of an element a in $\mathbb{Z}_{23}^\times$ is the smallest positive integer k such that $a^k \equiv 1 \pmod{23}$. To find the generators, we need to find the elements of order 22. The generators of $\mathbb{Z}_{23}^\times$ are the elements a such that $\gcd(a, 22) = 1$. The positive integers less than 22 that are coprime to 22 are 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, and 21. Therefore, the generators of $\mathbb{Z}_{23}^\times$ are 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, and 21. $\square$

Exercise 28.12. Is $x^3 + 2x + 3$ an irreducible polynomial of $\mathbb{Z}_5[x]$? Why? Express it as a product of irreducible polynomials $\mathbb{Z}_5[x]$.

Solution. Let $f(x) = x^3 + 2x + 3 \in \mathbb{Z}_5[x]$. Since $\mathbb{Z}_5$ is a field and $\deg f = 3$, f is reducible if and only if it has a root in $\mathbb{Z}_5$. We evaluate f at each element of $\mathbb{Z}_5$:

$$\begin{aligned} f(0) &= 3 \neq 0 \\ f(1) &= 1 + 2 + 3 = 6 \equiv 1 \pmod{5} \\ f(2) &= 8 + 4 + 3 = 15 \equiv 0 \pmod{5} \\ f(3) &= 27 + 6 + 3 = 36 \equiv 1 \pmod{5} \\ f(4) &= 64 + 8 + 3 = 75 \equiv 0 \pmod{5}. \end{aligned}$$

Thus $2, 4 \in \mathbb{Z}_5$ are roots of $f(x)$, and hence $f(x)$ is not irreducible in $\mathbb{Z}_5[x]$. Since $x - 2$ and $x - 4$ divide $f(x)$, we compute their product:

$$(x - 2)(x - 4) = x^2 - 6x + 8 \equiv x^2 - x + 3 \pmod{5}.$$

Dividing $f(x)$ by $x^2 - x + 3$, we obtain:

$$f(x) = (x^2 - x + 3)(x - 4).$$

Reducing all coefficients modulo 5, we have $x - 4 \equiv x + 1$, so the factorization becomes:

$$f(x) = (x^2 - x + 3)(x + 1).$$

Finally, note that $x^2 - x + 3 = (x + 1)(x + 3)$, in $\mathbb{Z}_5[x]$, so:

$$f(x) = (x + 1)^2(x + 3).$$

Each factor is linear, hence irreducible in $\mathbb{Z}_5[x]$, and this is the complete factorization of $f(x)$ into irreducible polynomials. $\square$

Exercise 28.36. (Remainder Theorem) Let $f(x) \in F[x]$ where F is a field, and let $\alpha \in F$. Show that the remainder $r(x)$ when $f(x)$ is divided by $x - \alpha$, in accordance with the division algorithm, is $f(\alpha)$.

Solution. Let $f(x) = q(x)(x - \alpha) + r(x)$, where $q(x), r(x) \in F[x]$ and $\deg r < 1$. Since $\deg r < 1$, $r(x)$ is a constant polynomial, say $r(x) = r$. Substituting $x = \alpha$ into the equation, we get:

$$f(\alpha) = q(\alpha)(\alpha - \alpha) + r = r.$$

Thus, the remainder r when $f(x)$ is divided by $x - \alpha$ is equal to $f(\alpha)$. $\square$

Exercise 30.2. Find all positive integers n such that $\mathbb{Z}_n$ contains a subring isomorphic to $\mathbb{Z}_2$.

Solution. Clearly, $n = 2$ works, since $\mathbb{Z}_2$ is isomorphic to itself. If $n > 2$, then $\mathbb{Z}_n$ contains a subring isomorphic to $\mathbb{Z}_2$ if and only if the following properties the identity element, e satisfies the following:

- (i) $e^2 \equiv e \pmod{n}$,
- (ii) $e + e \equiv 0 \pmod{n}$,
- (iii) and $e \neq 0$.

The second condition implies that $2e \equiv 0 \pmod{n}$, so n divides $2e$. Since $e \neq 0$, we have $e \geq 1$, and thus n divides 2. Therefore, we have $e = n/2$. Substituting $e = n/2$ into the first condition, we get:

$$\left(\frac{n}{2}\right)^2 \equiv \frac{n}{2} \pmod{n} \Rightarrow \frac{n^2}{4} - \frac{n}{2} = kn.$$

Since $n \neq 0$, we can divide both sides by n to obtain:

$$\frac{n-2}{4} = k.$$

For k to be an integer, $n-2$ must be divisible by 4, which means $n \equiv 2 \pmod{4}$. Therefore, the positive integers n such that $\mathbb{Z}_n$ contains a subring isomorphic to $\mathbb{Z}_2$ are those of the form $n = 4k+2$ for some integer $k \geq 0$. Lastly, we just check that $n = 4k+2$ works. Let $e = n/2 = 2k+1$. Then:

$$\begin{aligned} e^2 &\equiv (2k+1)^2 = 4k^2 + 4k + 1 \equiv 2k+1 \pmod{n} \\ e + e &\equiv 2(2k+1) = 4k+2 \equiv 0 \pmod{n}. \end{aligned}$$

Thus, the positive integers n such that $\mathbb{Z}_n$ contains a subring isomorphic to $\mathbb{Z}_2$ are those of the form $n = 4k+2$ for some integer $k \geq 0$. $\square$

Exercise 30.4. Give addition and multiplication tables for $2\mathbb{Z}/8\mathbb{Z}$. Are $2\mathbb{Z}/8\mathbb{Z}$ and $\mathbb{Z}_4$ isomorphic rings?

Solution. The elements of $2\mathbb{Z}/8\mathbb{Z}$ are $(0+8\mathbb{Z}), (2+8\mathbb{Z}), (4+8\mathbb{Z})$, and $6+8\mathbb{Z}$.

+	$0+8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$	$\times$	$0+8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$
$0+8\mathbb{Z}$	$0+8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$	$0+8\mathbb{Z}$	$0+8\mathbb{Z}$	$0+8\mathbb{Z}$	$0+8\mathbb{Z}$	$0+8\mathbb{Z}$
$2+8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$	$0+8\mathbb{Z}$	$2+8\mathbb{Z}$	$0+8\mathbb{Z}$	$4+8\mathbb{Z}$	$0+8\mathbb{Z}$	$4+8\mathbb{Z}$
$4+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$	$0+8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$	$0+8\mathbb{Z}$	$0+8\mathbb{Z}$	$0+8\mathbb{Z}$	$0+8\mathbb{Z}$
$6+8\mathbb{Z}$	$6+8\mathbb{Z}$	$0+8\mathbb{Z}$	$2+8\mathbb{Z}$	$4+8\mathbb{Z}$	$6+8\mathbb{Z}$	$0+8\mathbb{Z}$	$4+8\mathbb{Z}$	$0+8\mathbb{Z}$	$4+8\mathbb{Z}$

Since $\mathbb{Z}_4$ has unity but $2\mathbb{Z}/8\mathbb{Z}$ does not, these two rings are not isomorphic. $\square$

Exercise 30.17. Let $R = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$, and let R' consist of all 2×2 matrices of the form $\begin{bmatrix} a & 2b \\ b & a \end{bmatrix}$ for $a, b \in \mathbb{Z}$. Show that R is a subring of $\mathbb{R}$ and that R' is a subring of $M_2(\mathbb{Z})$. Then show that $\varphi : R \rightarrow R'$, where $\varphi(a + b\sqrt{2}) = \begin{bmatrix} a & 2b \\ b & a \end{bmatrix}$ is an isomorphism.

Solution. We first show that R is a subring of $\mathbb{R}$. Clearly, R is non-empty. Let $x = a + b\sqrt{2}$ and $y = c + d\sqrt{2}$ be elements of R , where $a, b, c, d \in \mathbb{Z}$. Then:

$$\begin{aligned} 3x + y &= (a + c) + (b + d)\sqrt{2} \in R \\ xy &= (a + b\sqrt{2})(c + d\sqrt{2}) = ac + 2bd + (ad + bc)\sqrt{2} \in R. \end{aligned}$$

Since R is closed under addition and multiplication, and contains the additive identity $0 = 0 + 0\sqrt{2}$, it is a subring of $\mathbb{R}$.

Next, we show that R' is a subring of $M_2(\mathbb{Z})$. Clearly, R' is non-empty. Let A and B be two elements of R' . Then:

$$A + B = \begin{bmatrix} a + c & 2(b + d) \\ b + d & a + c \end{bmatrix} \in R' \quad \text{and} \quad AB = \begin{bmatrix} ac + 2bd & 2(ad + bc) \\ ad + bc & ac + 2bd \end{bmatrix} \in R'.$$

Since R' is closed under addition and multiplication, and contains the additive identity $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, it is a subring of $M_2(\mathbb{Z})$.

Finally, we show that $\varphi : R \rightarrow R'$ defined by $\varphi(a + b\sqrt{2}) = \begin{bmatrix} a & 2b \\ b & a \end{bmatrix}$ is an isomorphism. We first show that φ is a homomorphism. Let $x = a + b\sqrt{2}$ and $y = c + d\sqrt{2}$ be elements of R . Then:

$$\begin{aligned} \varphi(x + y) &= \varphi((a + c) + (b + d)\sqrt{2}) = \begin{bmatrix} a + c & 2(b + d) \\ b + d & a + c \end{bmatrix} = \begin{bmatrix} a & 2b \\ b & a \end{bmatrix} + \begin{bmatrix} c & 2d \\ d & c \end{bmatrix} = \varphi(x) + \varphi(y) \\ \varphi(xy) &= \varphi((a + b\sqrt{2})(c + d\sqrt{2})) = \begin{bmatrix} ac + 2bd & 2(ad + bc) \\ ad + bc & ac + 2bd \end{bmatrix} = \begin{bmatrix} a & 2b \\ b & a \end{bmatrix} \begin{bmatrix} c & 2d \\ d & c \end{bmatrix} = \varphi(x)\varphi(y). \end{aligned}$$

Thus, φ is a homomorphism. Next, we show that φ is one-to-one. Suppose $\varphi(a + b\sqrt{2}) = \varphi(c + d\sqrt{2})$. Then:

$$\begin{bmatrix} a & 2b \\ b & a \end{bmatrix} = \begin{bmatrix} c & 2d \\ d & c \end{bmatrix}.$$

This implies that $a = c$ and $b = d$, so $a + b\sqrt{2} = c + d\sqrt{2}$. Thus, φ is one-to-one. Finally, we show that φ is onto. Let $A = \begin{bmatrix} a & 2b \\ b & a \end{bmatrix}$ be an element of R' . Then $\varphi(a + b\sqrt{2}) = A$, so φ is onto. Since φ is a homomorphism that is one-to-one and onto, it is an isomorphism. $\square$

Exercise 30.18. Show that each homomorphism from a field to a ring is either one-to-one or maps everything onto 0.

Solution. Let $\varphi : F \rightarrow R$ be a homomorphism from a field F to a ring R . Suppose φ is not one-to-one. Then there exist distinct elements $a, b \in F$ such that $\varphi(a) = \varphi(b)$. This implies that $\varphi(a - b) = \varphi(a) - \varphi(b) = 0$. Since F is a field, $a - b \neq 0$, so φ maps a non-zero element of F to 0. Let x be any element of F . Then:

$$\varphi(x) = \varphi\left(\frac{x}{a - b}(a - b)\right) = \varphi\left(\frac{x}{a - b}\right)\varphi(a - b) = \varphi\left(\frac{x}{a - b}\right) \cdot 0 = 0.$$

Thus, if φ is not one-to-one, it maps every element of F to 0.

Now, assume that φ doesn't map everything onto 0. Then there exists some element $c \in F$ such that $\varphi(c) \neq 0$. Since F is a field, c has a multiplicative inverse, say c^{-1} . Then:

$$\varphi(1) = \varphi(cc^{-1}) = \varphi(c)\varphi(c^{-1}) \neq 0.$$

Let x be any element of F . Then:

$$\varphi(x) = \varphi\left(\frac{x}{c}c\right) = \varphi\left(\frac{x}{c}\right)\varphi(c) \neq 0.$$

Thus, if φ doesn't map everything onto 0, it is one-to-one. Therefore, each homomorphism from a field to a ring is either one-to-one or maps everything onto 0. $\square$

Exercise 30.21. Let R and R' be rings and let $\varphi : R \rightarrow R'$ be a ring homomorphism such that $\varphi[R] \neq \{0'\}$. Show that if R has unity 1 and R' has no 0 divisors, then $\varphi(1)$ is a unity for R' .

Solution. Assume that R has unity 1 and R' has no zero divisors. Then we know that, for any nonzero $x, y \in R$, $\varphi(xy) = \varphi(x)\varphi(y) \neq 0$ if and only if $x \neq 0$ and $y \neq 0$. Then, if we let $x = 1 = y$, we obtain:

$$\varphi(1 \cdot 1) = \varphi(1)\varphi(1) = 1'\varphi(1) = \varphi(1)1' = \varphi(1).$$

Thus, $\varphi(1)$ is a unity for R' . $\square$

Exercise 30.22. Let $\varphi : R \rightarrow R'$ be a ring homomorphism and let N be an ideal of R .

(i) Show that $\varphi[N]$ is an ideal of $\varphi[R]$.

(ii) Give an example to show that $\varphi[N]$ need not be an ideal of R' .

(iii) Let N' be an ideal of either $\varphi[R]$ or of R' . Show that $\varphi^{-1}[N']$ is an ideal of R .

Solution to (i). Let $I = \varphi[N]$. Since N is an ideal of R , it is non-empty, so I is non-empty. Let $x, y \in I$. Then there exist $a, b \in N$ such that $\varphi(a) = x$ and $\varphi(b) = y$. Since N is an ideal, $a - b \in N$, so $\varphi(a - b) = \varphi(a) - \varphi(b) = x - y \in I$. Let $r \in \varphi[R]$ and let $x \in I$. Then there exists $a \in N$ such that $\varphi(a) = x$. Since $\varphi[R]$ is the image of φ , there exists some element $s \in R$ such that $\varphi(s) = r$. Since N is an ideal, $sa \in N$, so $\varphi(sa) = \varphi(s)\varphi(a) = rx \in I$. Similarly, since $as \in N$, we have $\varphi(as) = \varphi(a)\varphi(s) = xr \in I$. Thus, I is an ideal of $\varphi[R]$. $\square$

Solution to (ii). Let $R = \mathbb{Z}$ and $R' = \mathbb{Q}$, with $f : \mathbb{Z} \rightarrow \mathbb{Q}$. Let $I = 3\mathbb{Z}$. Suppose that $f(I)$ is an ideal of $\mathbb{Q}$. Then $1/3 \in \mathbb{Q}$, but $1/3 \notin f(I)$, since $f(I)$ consists of all rational numbers of the form $3k$ for some integer k . Thus, $f(I)$ is not an ideal of $\mathbb{Q}$. $\square$

Solution to (iii). Let $I = \varphi^{-1}[N']$. Since N' is an ideal of either $\varphi[R]$ or R' , it is non-empty, so I is non-empty. Let $x, y \in I$. Then $\varphi(x), \varphi(y) \in N'$, so $\varphi(x) - \varphi(y) = \varphi(x - y) \in N'$. Thus, $x - y \in I$. Let $r \in R$ and let $x \in I$. Then $\varphi(r) \in \varphi[R]$ and $\varphi(x) \in N'$, so $\varphi(r)\varphi(x) = \varphi(rx) \in N'$. Similarly, since $\varphi(x)\varphi(r) = \varphi(xr) \in N'$, we have $xr \in I$. Thus, I is an ideal of R . $\square$

Exercise 30.30. An element a of a ring R is *nilpotent* if $a^n = 0$ for some $n \in \mathbb{Z}^+$. Show that the collection of all nilpotent elements in a commutative ring R is an ideal, the *nilradical* of R .

Solution. Let $I = \{a \in R \mid a^n = 0 \text{ for some } n \in \mathbb{Z}^+\}$. We first show that I is non-empty. Since $0^1 = 0$, we have $0 \in I$, so I is non-empty. Let $x, y \in I$. Then there exist positive integers m and n such that $x^m = 0$ and $y^n = 0$. Since R is commutative, we have:

$$(x - y)^{m+n} = \sum_{k=0}^{m+n} \binom{m+n}{k} x^k (-y)^{m+n-k} = \sum_{k=0}^{m-1} \binom{m+n}{k} x^k (-y)^{m+n-k} + \sum_{k=m}^{m+n} \binom{m+n}{k} x^k (-y)^{m+n-k}$$

$$\begin{aligned}
&= \sum_{k=0}^{m-1} \binom{m+n}{k} x^k (-y)^{m+n-k} + \sum_{k=m}^{m+n} \binom{m+n}{k} 0 \cdot (-y)^{m+n-k} = \sum_{k=0}^{m-1} \binom{m+n}{k} x^k (-y)^{m+n-k} \\
&= \sum_{k=0}^{m-1} \binom{m+n}{k} 0 \cdot (-y)^{m+n-k} = 0.
\end{aligned}$$

Thus, $x - y \in I$. Let $r \in R$ and let $x \in I$. Then there exists a positive integer m such that $x^m = 0$. Since R is commutative, we have:

$$(rx)^m = r^m x^m = r^m \cdot 0 = 0 \quad \text{and} \quad (xr)^m = x^m r^m = 0 \cdot r^m = 0.$$

Thus, $rx \in I$ and $xr \in I$. Therefore, I is an ideal of R , and it is called the nilradical of R . □